ZAHIR JACOB

ARTIFICIAL INTELLIGENCE STUDENT | DATA SCIENCE & MACHINE LEARNING

Rosario, Argentina | +54 9 341 608-1343 | zahirjacob81@gmail.com | LinkedIn | GitHub

PROFILE

Artificial Intelligence student at the National University of Rosario (UNR), focused on Data Science and Machine Learning. Hands-on experience with Python, Pandas, NumPy, Scikit-learn, SQL and Jupyter through end-to-end projects involving exploratory data analysis, data preprocessing, predictive modeling, model evaluation and visualization. Interested in applying data-driven approaches to real-world problems and building reproducible machine learning solutions.

EDUCATION

University Technical Degree in Artificial Intelligence - FCEIA, National University of Rosario (UNR) **2023 - Present**

Final semester. Relevant areas: Machine Learning, Data Mining, Databases, Business Intelligence, Computer Vision and NLP.

TECHNICAL SKILLS

Data Science & Programming: Python, Pandas, NumPy, Jupyter Notebook, Google Colab, exploratory data analysis (EDA), data preprocessing, feature engineering, Matplotlib.
Machine Learning: Scikit-learn, classification, model selection, stratified cross-validation, hyperparameter optimization, SMOTE, model evaluation (F1, precision, recall, PR-AUC, ROC-AUC).
Databases & Visualization: SQL, SQL Server, SSMS, SSIS, ETL, data warehousing, dimensional modeling, Power BI.
Tools & Additional AI: Git, GitHub, Docker, Visual Studio Code; NLP, RAG, ChromaDB, BM25, OpenCV.

SELECTED PROJECTS

Australia Rain Prediction & Live ML Application | *Python, Pandas, NumPy, Scikit-learn, Docker* GitHub · Live app

- Built an end-to-end classification pipeline on 142,153 labeled weather observations from 49 Australian locations, including EDA, preprocessing, feature engineering and leakage-aware model evaluation.
- Compared 23 candidate configurations with 5-fold stratified cross-validation and complementary expanding-window temporal evaluation; selected a neural network achieving F1 0.640 on the temporal check and ROC-AUC 0.885 on the held-out test.
- Extended the academic model into a deployed web app that fetches real weather observations, performs reproducible preprocessing and runs TensorFlow-free neural-network inference in NumPy.

Data Warehouse & Sales Dashboard | *SQL Server, SSIS, SSMS, Power BI*

- Integrated multiple data sources through ETL processes; built dimensions, fact tables, views and stored procedures, then developed a Power BI dashboard for sales, customer, category, location and Pareto analysis.

Multi-source RAG Assistant | *Python, Pandas, Scikit-learn, ChromaDB, BM25* GitHub

- Built a conversational assistant over documents, tabular data and a knowledge graph using intent classification, hybrid semantic/BM25 retrieval, cross-encoder re-ranking and a ReAct agent; the adopted local-LLM intent classifier reached 0.96 macro F1 on the project test set.

LANGUAGES

Spanish: Native. **English:** Upper-intermediate; advanced comprehension and functional spoken communication.